MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DIODE, PROGRAMMABLE, DIGITAL, KEY CODE ABE

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic programmable diode module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specificed in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.052 failures per million operating hours for class I or 0.094 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module confirguration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be ABE.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirement for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/22-1 for class I and M28787/22-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

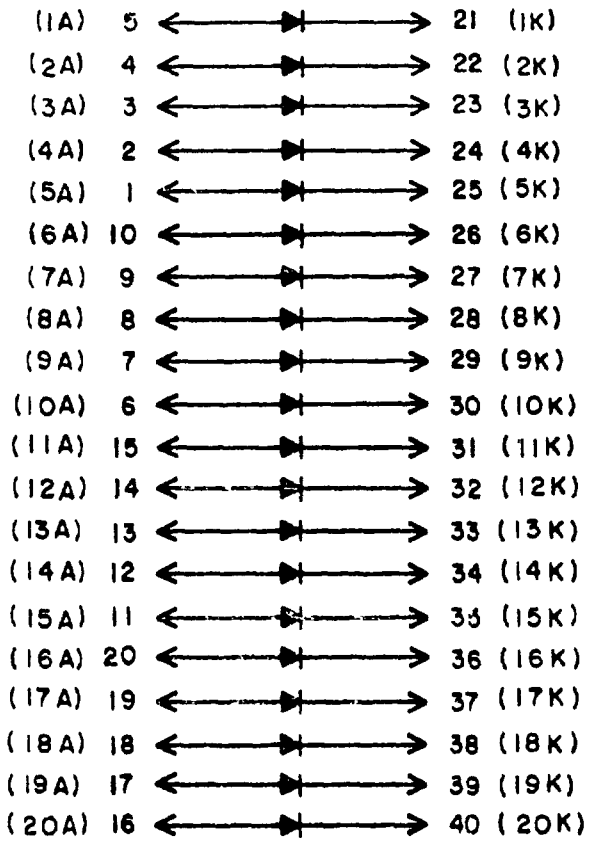
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. 1A-20A = Anode.
2. 1K-20K = Cathode.

FIGURE 1. ABE module function diagram, programmable diode.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/ 2/	25° ±5°C 3/		Class I 3/				Class II 3/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	II	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	X	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B19	Integrity													
	Initial	III	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	X	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
C1-C20	Reverse current													
	Initial	IV	X	0.0	0.1	0.0	2.5	0.0	2.5	0.0	2.5	0.0	2.5	µA
	End-of-life	IV	X	0.0	1.0	0.0	5.0	0.0	5.0	0.0	5.0	0.0	5.0	µA
D1-D20	Foward voltage drop (I_F = 0.25 mA)													
	Initial	V	X	460	640	350	710	350	710	300	860	300	800	mV
	End-of-life	V	X	440	660	310	730	310	730	280	880	280	820	mV
E1-E20	Foward voltage drop (I_F = 1.0 mA)													
	Initial	V	X	510	700	400	770	400	770	300	900	350	860	mV
	End-of-life	V	X	490	720	360	790	360	790	280	920	330	880	mV
F1-F20	Foward voltage drop (I_F = 20 mA)													
	Initial	V	X	670	860	580	910	580	910	500	1030	540	990	mV
	End-of-life	V	X	650	880	550	930	550	930	480	1050	520	1010	mV
G1-G20	Foward voltage drop (I_F = 300 mA; 30 µs pulse, 3% duty cycle)													
	Initial	VI	S	0.90	1.20	0.85	1.24	0.85	1.24	0.78	1.32	0.81	1.29	V
	End-of-life	VI	S	0.90	1.22	0.83	1.26	0.83	1.26	0.76	1.34	0.79	1.31	V
H1-H20	Reverse recovery time													
	Initial	VII	S	1	8	1	8	1	8	1	9	1	9	ns
	End-of-life	VII	S	1	10	1	10	1	10	1	10	1	10	ns

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ 2/	25° ±5°C 3/		Class I 3/				Class II 3/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
J1-J20	Capacitance													
	Initial	VIII	S	1	8	1	8	1	8	1	8	1	8	pF
	End-of-life	VIII	S	1	10	1	10	1	10	1	10	1	10	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.c Order of testing. Test conditions A through J shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through J in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through G shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be nonoperating. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests A through J shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through J shall be performed in the order specified in 4.4.2 using the 25°±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Cross reference of classification. Military Specification MIL-M-28787/22 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658626. Part number M28787/22-1 supersedes part numbers 2658626 and M28787/22.

6.5 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 85°C end-of-life parameters from table I.

	Class I	Class 2	Unit
Power dissipation	1.25 max	1.25 max	W
Diode continuous forward current	135 max	135 max	mA
Diode reverse voltage	50 max	50 max	V
Reverse recovery time	10 max	10 max	ns

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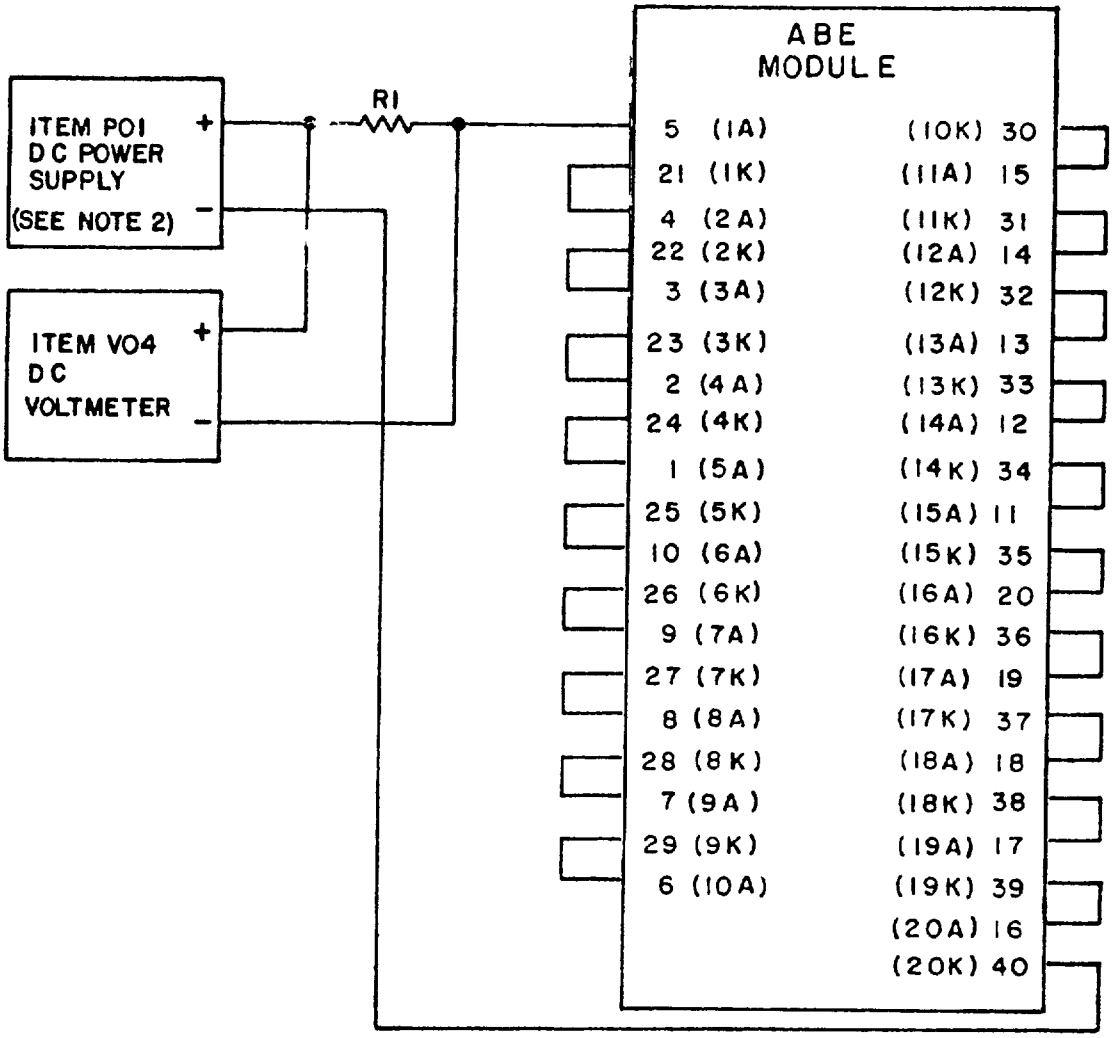
6.6 Changes from previous issue. Asterisks are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - EC

(Project 5963-N359)

Agent:
DLA - ES

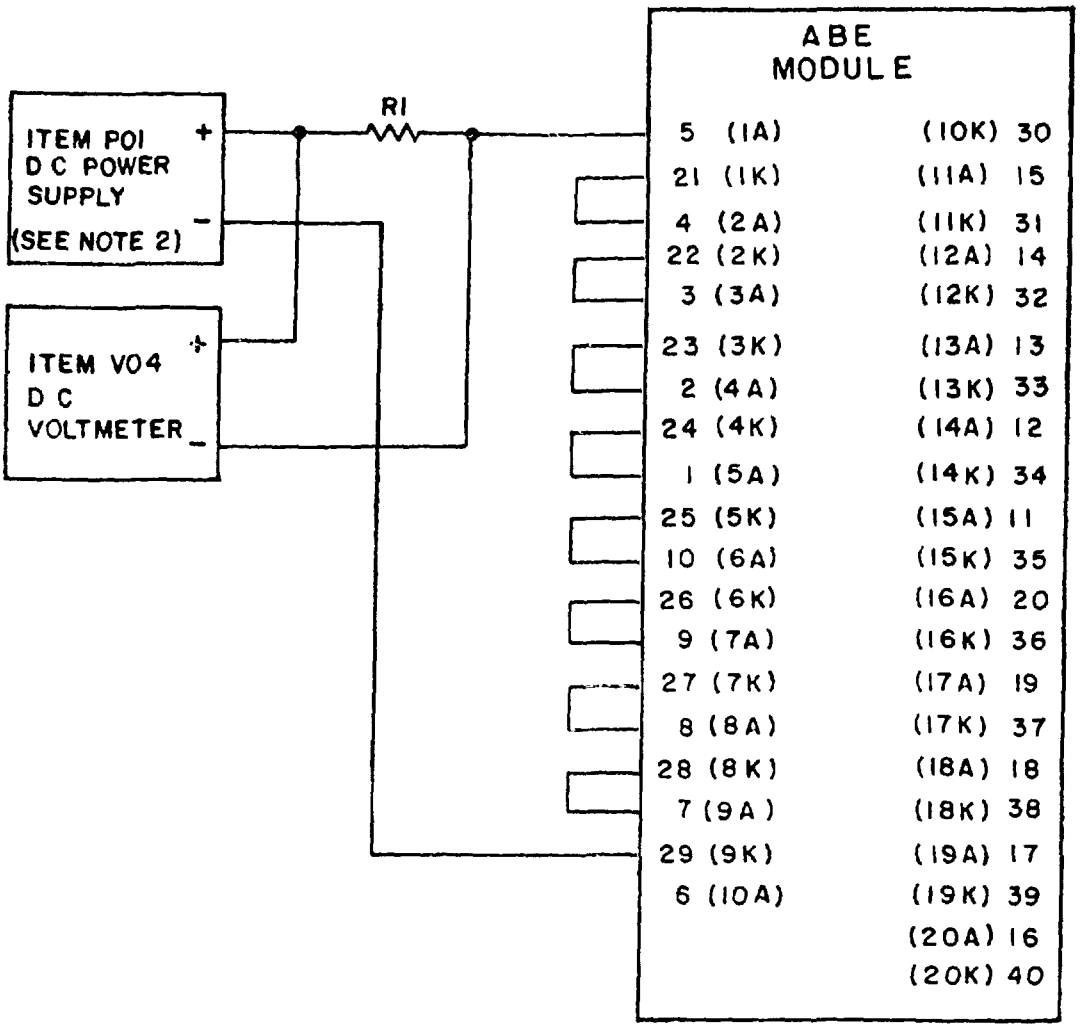
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NOTES:
1. R1 = 76.8Ω ±1%, 1/2 W.
2. Adjust item P01 for a voltage reading of 5.00 V ±1% across R1 as read on item V04 and corresponding to a current of 65 mA

FIGURE 2. Operating temperature and life test circuit.

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NOTES
1. R1 = 40Ω ±1%, 1 W.
2. Adjust item P01 for a voltage reading of 5.40 V ±1% across R1 as read on item V04 and corresponding to a current of 135 mA.

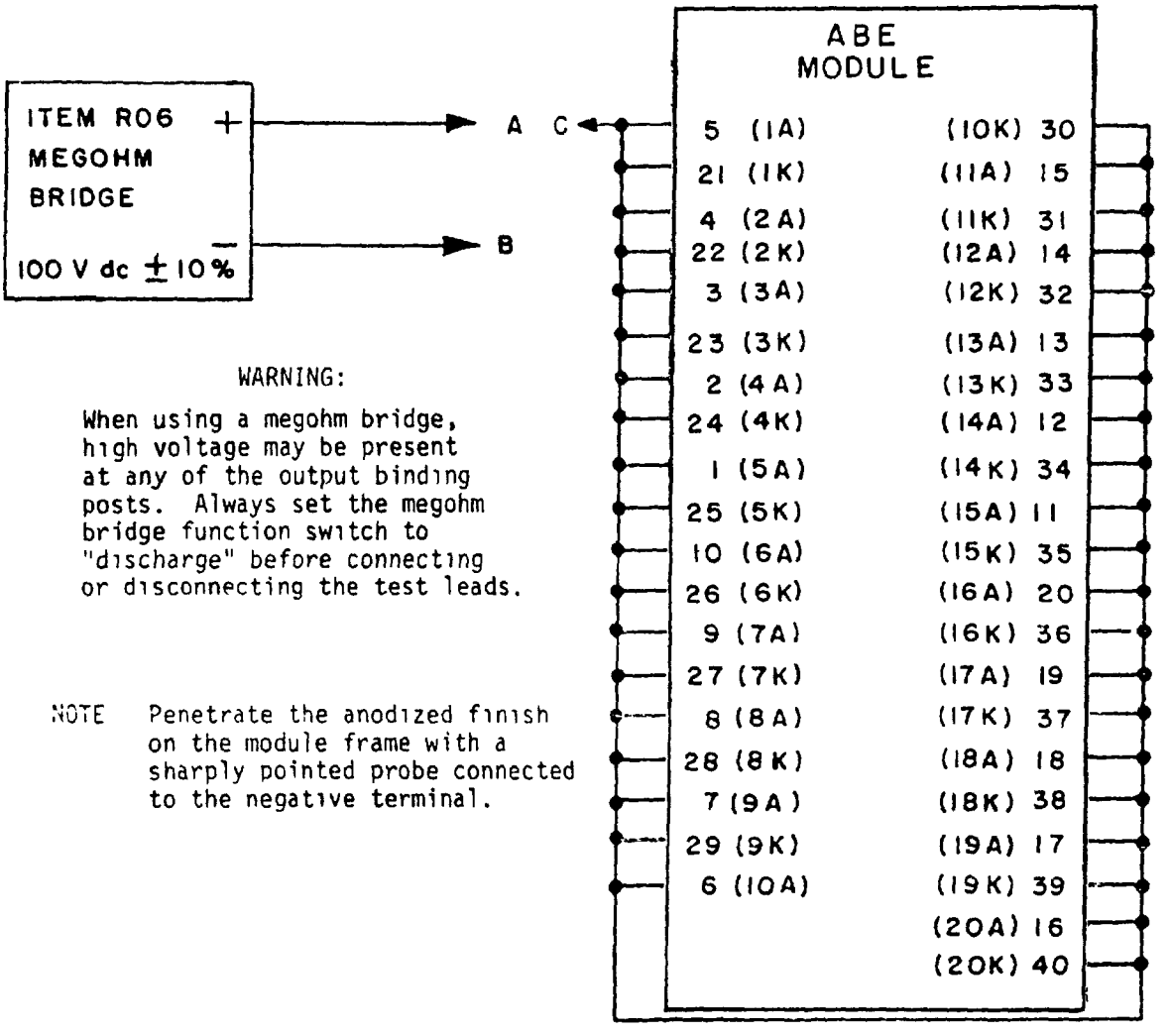
FIGURE 3. Component temperature test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	4

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

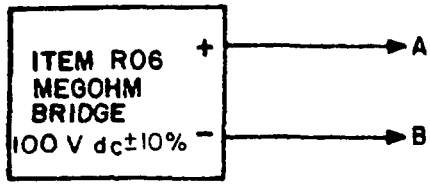
FIGURE 4. Isolation test circuit for table II.

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TABLE III. Integrity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Integrity	5, 21	All pins except 5, 21	10	∞	MΩ	5
B2		4, 22	4, 22				
B3		3, 23	3, 23				
B4		2, 24	2, 24				
B5		1, 25	1, 25				
B6		10, 26	10, 26				
B7		9, 27	9, 27				
B8		8, 28	8, 28				
B9		7, 29	7, 29				
B10		6, 30	6, 30				
B11		15, 31	15, 31				
B12		14, 32	14, 32				
B13		13, 33	13, 33				
B14		12, 34	12, 34				
B15		11, 35	11, 35				
B16		20, 36	20, 36				
B17		19, 37	19, 37				
B18		16, 38	18, 38				
B19		17, 39	17, 39				

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

ABE MODULE	
5 (1A)	(10K) 30
21 (1K)	(11A) 15
4 (2A)	(11K) 31
22 (2K)	(12A) 14
3 (3A)	(12K) 32
23 (3K)	(13A) 13
2 (4A)	(13K) 33
24 (4K)	(14A) 12
1 (5A)	(14K) 34
25 (5K)	(15A) 11
10 (6A)	(15K) 35
26 (6K)	(16A) 20
9 (7A)	(16K) 36
27 (7K)	(17A) 19
8 (8A)	(17K) 37
28 (8K)	(18A) 18
7 (9A)	(18K) 38
29 (9K)	(19A) 17
6 (10A)	(19K) 39
	(20A) 16
	(20K) 40

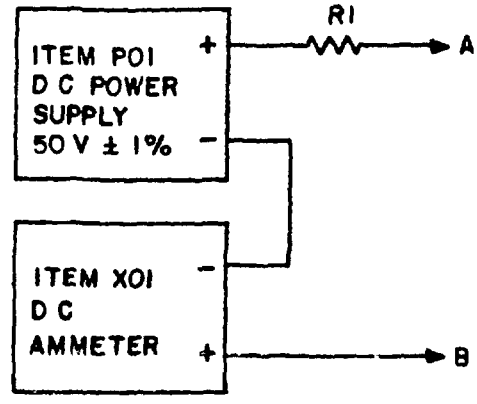
FIGURE 5. Integrity test circuit for table III.

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TABLE IV. Reverse current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Reverse current	21	5	0.0	0.1	µA	6
C2		22	4				
C3		23	3				
C4		24	2				
C5		25	1				
C6		26	10				
C7		27	9				
C8		28	8				
C9		29	7				
C10		30	6				
C11		31	15				
C12		32	14				
C13		33	13				
C14*		34	12				
C15		35	11				
C16		36	20				
C17		37	19				
C18		38	18				
C19		39	17				
C20		40	16				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: R1 = 1 MΩ ±1%, 1/8 W.

ABE
MODULE

5 (1A)	(10K) 30
21 (1K)	(11A) 15
4 (2A)	(11K) 31
22 (2K)	(12A) 14
3 (3A)	(12K) 32
23 (3K)	(13A) 13
2 (4A)	(13K) 33
24 (4K)	(14A) 12
1 (5A)	(14K) 34
25 (5K)	(15A) 11
10 (6A)	(15K) 35
26 (6K)	(16A) 20
9 (7A)	(16K) 36
27 (7K)	(17A) 19
8 (8A)	(17K) 37
28 (8K)	(18A) 18
7 (9A)	(18K) 38
29 (9K)	(19A) 17
6 (10A)	(19K) 39
	(20A) 16
	(20K) 40

FIGURE 6. Reverse current test circuit for table IV.

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TABLE V. Forward voltage drop test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
D1	Forward voltage drop (I_F = 0.25 mA)	5	Open	Open	5	21	460	640	mV	7
D2		4			4	22				
D3		3			3	23				
D4		2			2	24				
D5		1			1	25				
D6		10			10	26				
D7		9			9	27				
D8		8			8	28				
D9		7			7	29				
D10		6			6	30				
D11		15			15	31				
D12		14			14	32				
D13		13			13	33				
D14		12			12	34				
D15		11			11	35				
D16		20			20	36				
D17		19			19	37				
D18		18			18	38				

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TABLE V. Forward voltage drop test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
D19	Forward voltage drop (I_F = 0.25 mA)	17	Open	Open	17	39	460	640	mV	7
D20	Forward voltage drop (I_F = 0.25 mA)	16	Open	Open	16	40	460	640		
E1	Forward voltage drop (I_F = 1.0 mA)	Open	5		5	21	510	700		
E2			4		4	22				
E3			3		3	23				
E4			2		2	24				
E5			1		1	25				
E6			10		10	26				
E7			9		9	27				
E8			8		8	28				
E9			7		7	29				
E10			6		6	30				
E11			15		15	31				
E12			14		14	32				
E13			13		13	33				
E14			12		12	34				
E15			11		11	35				

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TABLE V. Forward voltage drop test - Continued.

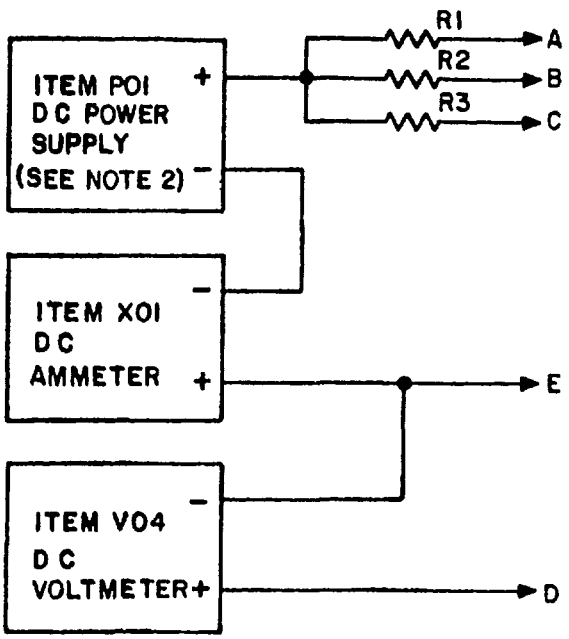
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E16	Forward voltage drop (I_F = 1.0 mA)	Open	20	Open	20	36	510	700	mV	7
E17			19		19	37				
E18			18		18	38				
E19			17		17	39				
E20			16		16	40				
F1	Foward voltage drop (I_F = 20.0 mA)		Open	5	5	21	670	860		
F2				4	4	22				
F3				3	3	23				
F4				2	2	24				
F5				1	1	25				
F6				10	10	26				
F7				9	9	27				
F8				8	8	28				
F9				7	7	29				
F10				6	6	30				
F11				15	15	31				
F12				14	14	32				

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TABLE V. Forward voltage drop test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F13	Forward voltage drop (I_F = 20.0 mA)	Open	Open	13	13	33	670	860	mV	7
F14	↓	↓	↓	12	12	34	↓	↓	↓	↓
F15				11	11	35				
F16				20	20	36				
F17				19	19	37				
F18				18	18	38				
F19				17	17	39				
F20				16	16	40				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

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24 (4K)	(14A) 12
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25 (5K)	(15A) 11
10 (6A)	(15K) 35
26 (6K)	(16A) 20
9 (7A)	(16K) 36
27 (7K)	(17A) 19
8 (8A)	(17K) 37
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	(20A) 16
	(20K) 40

NOTES:
1. R1 = 20 kΩ ±1%, 1/8 W.
   R2 = 4.99 kΩ ±1%, 1/8 W.
   R3 = 249Ω ±1%, 1/4 W.
2. Test conditions:
   a. For test conditions D1-D20 adjust supply voltage to obtain a forward current of 0.25 ± 0.0025 mA.
   b. For test conditions E1-E20 adjust supply voltage to obtain a forward current of 1.0 ± 0.01 mA.
   c. For test conditions F1-F20 adjust supply voltage to obtain a forward current of 20.0 ±0.2 mA.

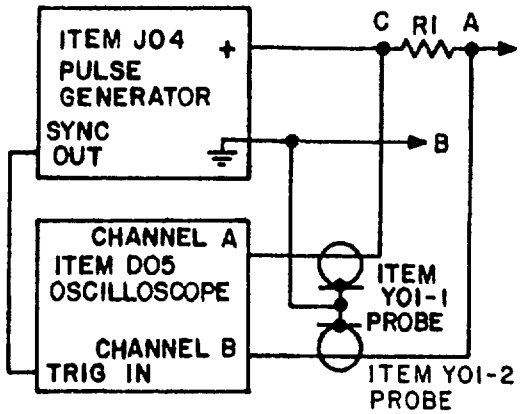
FIGURE 7. Forward voltage drop test circuit for table V.

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TABLE VI. Forward voltage drop test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	Foward voltage drop (I_F = 300 mA)	5	21	0.90	1.20	V	8,9
G2		4	22				
G3		3	23				
G4		2	24				
G5		1	25				
G6		10	26				
G7		9	27				
G8		8	28				
G9		7	29				
G10		6	30				
G11		15	31				
G12		14	32				
G13		13	33				
G14		12	34				
G15		11	35				
G16		20	36				
G17		19	37				
G18		18	38				
G19		17	39				
G20		16	40				

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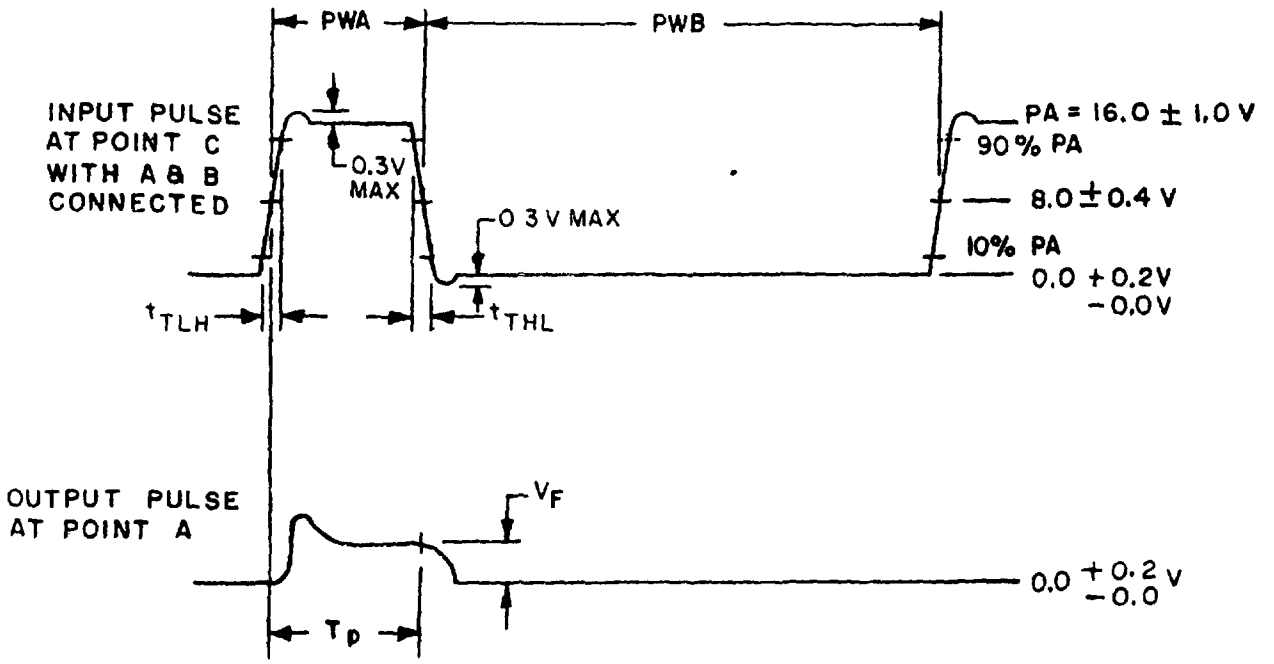


NOTE: R1 = 49.9Ω ±1%, 1/2 W.

ABE MODULE	
5 (1A)	(10K) 30
21 (1K)	(11A) 15
4 (2A)	(11K) 31
22 (2K)	(12A) 14
3 (3A)	(12K) 32
23 (3K)	(13A) 13
2 (4A)	(13K) 33
24 (4K)	(14A) 12
1 (5A)	(14K) 34
25 (5K)	(15A) 11
10 (6A)	(15K) 35
26 (6K)	(16A) 20
9 (7A)	(16K) 36
27 (7K)	(17A) 19
8 (8A)	(17K) 37
28 (8K)	(18A) 18
7 (9A)	(18K) 38
29 (9K)	(19A) 17
6 (10A)	(19K) 39
	(20A) 16
	(20K) 40

FIGURE 8. Forward voltage drop test circuit for table VI.

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NOTE: PWA = 30 ±2 µs.
PWB = 970 ±50 µs.
$t_{TLH} = t_{THL} = 0.5 \pm 0.1$ µs.
$T_p = 25 \pm 2$ µs.

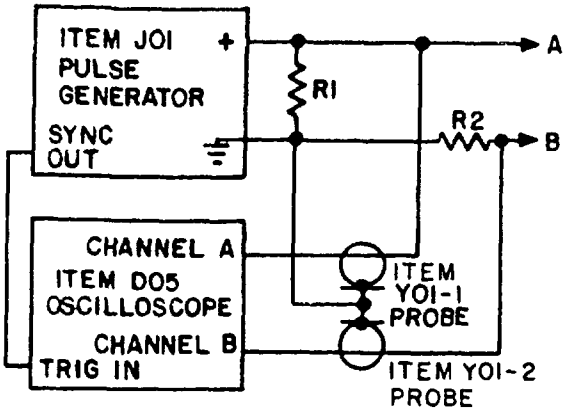
FIGURE 9. Forward voltage drop test circuit waveforms for table VI.

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TABLE VII. Reverse recovery time test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Reverse recovery time	5	21	1	8	ns	10,11
H2		4	22				
H3		3	23				
H4		2	24				
H5		1	25				
H6		10	26				
H7		9	27				
H8		8	28				
H9		7	29				
H10		6	30				
H11		15	31				
H12		14	32				
H13		13	33				
H14		12	34				
H15		11	35				
H16		20	36				
H17		19	37				
H18		18	38				
H19		17	39				
H20		16	40				

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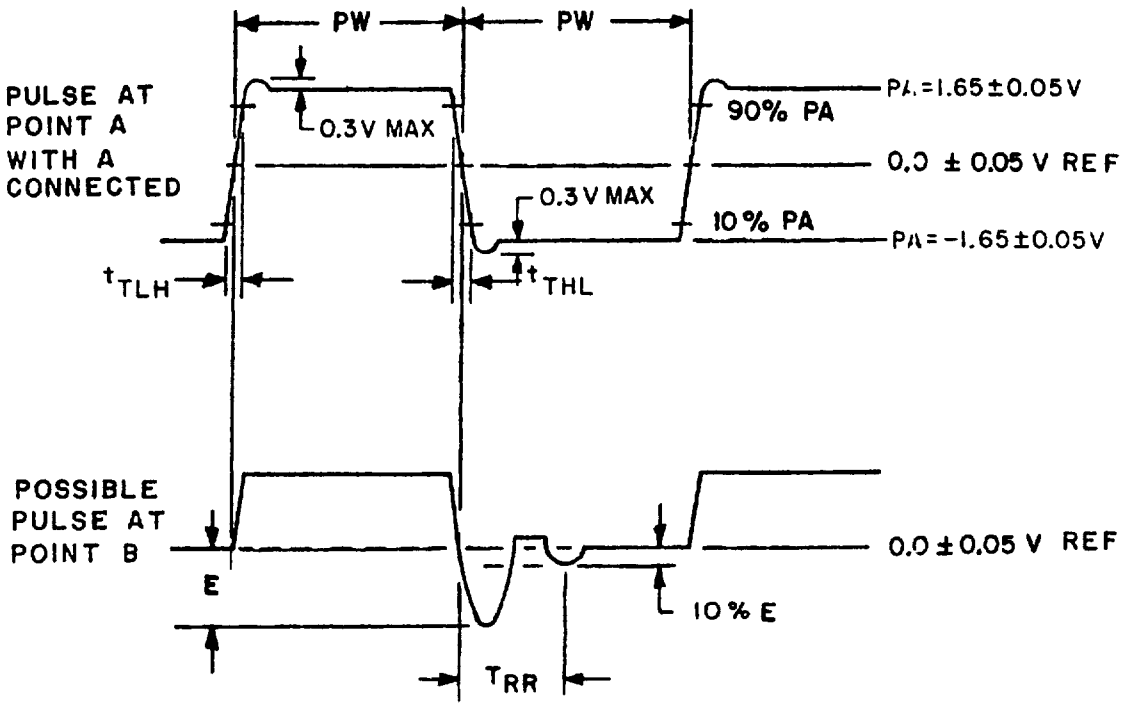


NOTES.
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 100Ω ±1%, 1/8 W.

ABE MODULE	
5 (1A)	(10K) 30
21 (1K)	(11A) 15
4 (2A)	(11K) 31
22 (2K)	(12A) 14
3 (3A)	(12K) 32
23 (3K)	(13A) 13
2 (4A)	(13K) 33
24 (4K)	(14A) 12
1 (5A)	(14K) 34
25 (5K)	(15A) 11
10 (6A)	(15K) 35
26 (6K)	(16A) 20
9 (7A)	(16K) 36
27 (7K)	(17A) 19
8 (8A)	(17K) 37
28 (8K)	(18A) 18
7 (9A)	(18K) 38
29 (9K)	(19A) 17
6 (10A)	(19K) 39
	(20A) 16
	(20K) 40

FIGURE 10. Reverse recovery time test circuit for table VII.

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NOTES:
1. Input pulse parameters:
   PW = 1.0 ±0.1 µs.
   $t_{TLH} = t_{THL} = 3 \pm 1$ ns.
2. Output pulse parameters:
   T_{RR} = Reverse recovery time.

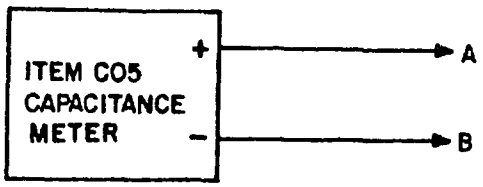
FIGURE 11 Reverse recovery time test circuit waveforms for table VII.

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TABLE VIII. Capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Capacitance	21	5	1	8	pF	12
J2		22	4				
J3		23	3				
J4		24	2				
J5		25	1				
J6		26	10				
J7		27	9				
J8		28	8				
J9		29	7				
J10		30	6				
J11		31	15				
J12		32	14				
J13		33	13				
J14		34	12				
J15		35	11				
J16		36	20				
J17		37	19				
J18		38	18				
J19		39	17				
J20		40	16				

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ABE MODULE	
5 (1A)	(10K) 30
21 (1K)	(11A) 15
4 (2A)	(11K) 31
22 (2K)	(12A) 14
3 (3A)	(12K) 32
23 (3K)	(13A) 13
2 (4A)	(13K) 33
24 (4K)	(14A) 12
1 (5A)	(14K) 34
25 (5K)	(15A) 11
10 (6A)	(15K) 35
26 (6K)	(16A) 20
9 (7A)	(16K) 36
27 (7K)	(17A) 19
8 (8A)	(17K) 37
28 (8K)	(18A) 18
7 (9A)	(18K) 38
29 (9K)	(19A) 17
6 (10A)	(19K) 39
	(20A) 16
	(20K) 40

NOTES:
1. Do not attach test connector or wires to any of the module pins other than those indicated in table VII. Stray capacitance added by other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

FIGURE 12. Capacitance test circuit for table VIII.

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